

# Laporan Analisis Parameter Kisi XRD (Struktur Kubus)

di analisis ke 1 - 3 itu pr , dan yang ke empat itu quiz.

- denga menggunakan persamaan : parameter kisi  $= \lambda / 2 \sin \theta \sqrt{h^2 + k^2 + l^2}$
- perhitungan detail = <https://github.com/muhammadsyaddad/fisdat>
- dengan  $d (\text{\AA}) = \lambda / 2 \sin \theta$
- dan  $a (\text{\AA}) = d (\text{\AA}) * \sqrt{h^2 + k^2 + l^2}$
- dan  $d (\text{\AA})$  adalah standar deviasi
- kemudian a itu paremter kisi dan di avarage (rata rata)

## Analisis Struktur Kisi: CsCl

| 2theta  | h | k | l | d (Å) | a (Å)  |
|---------|---|---|---|-------|--------|
| 21.550° | 1 | 0 | 0 | 4.12  | 4.12   |
| 30.622° | 1 | 1 | 0 | 2.917 | 4.1252 |
| 37.766° | 1 | 1 | 1 | 2.38  | 4.1223 |
| 43.869° | 2 | 0 | 0 | 2.062 | 4.124  |
| 49.380° | 2 | 1 | 0 | 1.844 | 4.1233 |
| 54.473° | 2 | 1 | 1 | 1.683 | 4.1225 |
| 63.829° | 2 | 2 | 0 | 1.457 | 4.121  |
| 68.193° | 3 | 0 | 0 | 1.374 | 4.122  |
| 72.411° | 3 | 1 | 0 | 1.304 | 4.1236 |
| 76.584° | 3 | 1 | 1 | 1.243 | 4.1226 |

- Nilai Parameter Kisi Rata-Rata (a): 4.1226 Å
  - Standar Deviasi: 1.48e-03 Å
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## Analisis Struktur Kisi: Al<sub>3</sub>Ti<sub>5</sub>O<sub>2</sub>

| 2theta  | h | k | l | d (Å) | a (Å)  |
|---------|---|---|---|-------|--------|
| 22.451° | 1 | 1 | 1 | 3.96  | 6.8589 |
| 29.086° | 2 | 1 | 0 | 3.07  | 6.8648 |
| 39.344° | 3 | 0 | 0 | 2.29  | 6.87   |
| 41.418° | 3 | 1 | 0 | 2.18  | 6.8938 |
| 43.508° | 3 | 1 | 1 | 2.08  | 6.8986 |
| 49.481° | 3 | 2 | 1 | 1.842 | 6.8922 |
| 56.451° | 3 | 3 | 0 | 1.63  | 6.9155 |
| 59.949° | 4 | 2 | 0 | 1.543 | 6.9005 |
| 61.579° | 4 | 2 | 1 | 1.506 | 6.9014 |
| 63.211° | 3 | 3 | 2 | 1.471 | 6.8996 |
| 67.865° | 4 | 3 | 0 | 1.381 | 6.905  |
| 69.410° | 5 | 1 | 0 | 1.354 | 6.9041 |

• Nilai Parameter Kisi Rata-Rata (a): 6.8920 Å

• Standar Deviasi: 1.77e-02 Å

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## Analisis Struktur Kisi: Ba<sub>0.5</sub>Sr<sub>0.5</sub>TiO<sub>3</sub>

| 2theta  | h | k | l | d (Å)  | a (Å)  |
|---------|---|---|---|--------|--------|
| 22.494° | 1 | 0 | 0 | 3.9495 | 3.9495 |
| 32.032° | 1 | 1 | 0 | 2.7919 | 3.9483 |
| 39.498° | 1 | 1 | 1 | 2.2797 | 3.9485 |
| 45.944° | 2 | 0 | 0 | 1.9737 | 3.9474 |
| 51.755° | 2 | 1 | 0 | 1.7649 | 3.9465 |
| 57.114° | 2 | 1 | 1 | 1.6114 | 3.9471 |
| 67.011° | 2 | 2 | 0 | 1.3954 | 3.9469 |
| 71.674° | 3 | 0 | 0 | 1.3157 | 3.947  |

| 2theta   | h | k | l | d (Å)  | a (Å)  |
|----------|---|---|---|--------|--------|
| 76.225°  | 3 | 1 | 0 | 1.248  | 3.9466 |
| 80.679°  | 3 | 1 | 1 | 1.19   | 3.9468 |
| 85.052°  | 2 | 2 | 2 | 1.1396 | 3.9478 |
| 89.459°  | 3 | 2 | 0 | 1.0945 | 3.9464 |
| 93.808°  | 3 | 2 | 1 | 1.0549 | 3.9471 |
| 102.630° | 4 | 0 | 0 | 0.9868 | 3.9472 |
| 107.130° | 4 | 1 | 0 | 0.9575 | 3.9477 |

- Nilai Parameter Kisi Rata-Rata (a): 3.9474 Å
- Standar Deviasi: 8.41e-04 Å

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## Analysis data (quiz)

| 2theta  | h | k | l | d (Å)  | a (Å)   |
|---------|---|---|---|--------|---------|
| 22.000° | 2 | 1 | 1 | 4.0368 | 9.888   |
| 24.000° | 2 | 2 | 0 | 3.7047 | 10.4785 |
| 29.000° | 2 | 2 | 2 | 3.0763 | 10.6567 |
| 33.000° | 1 | 2 | 3 | 2.712  | 10.1474 |
| 34.000° | 4 | 0 | 0 | 2.6345 | 10.538  |

- Nilai Parameter Kisi Rata-Rata (a): 10.3417 Å
  - Standar Deviasi: 3.16e-01 Å
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